Jay Ullaskumar Trivedi

AI · ML Researcher

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Jay Trivedi
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Summary

Research focused M.Tech student in Artificial Intelligence with a strong foundation in Computer Engineering, specializing in machine learning, deep learning, and reinforcement learning. Experienced in designing and developing novel architectures for autonomous systems, AI models for audio classification and time-series forecasting.Proficient in Python with hands-on experience in PyTorch, TensorFlow, and CUDA. Building and optimizing AI models for real-world applications using advanced techniques.

Skills

Languages: Python, PHP Laravel
Domains: Machine Learning, Reinforcement Learning, Deep Learning, Computer Vision, Audio Classification
Frameworks: PyTorch, NumPy, Pandas, Matplotlib, Stable-Baselines3
Simulator: MetaDrive
Specialized Areas: Autonomous Driving, Bioacoustic AI, Time Series Forecasting, Real-Time Inference Systems

Experience

Intern - Live Bird Species Recognition System (AI-Powered) — *June 2025 - July 2025*
DIAT-DRDO, Pune

Intern - Python Django — *June 2022 - July 2022*
Creart, Ahmedabad

Education

Master of Technology, Artificial Intelligence — *2024 - 2026*
Pandit Deendayal Energy University (PDEU), Gandhinagar

Bachelor of Technology, Computer Engineering — *2019 - 2023*
Silver Oak College of Engineering and Technology, Ahmedabad

Higher Secondary (12th, Science) — *2018 - 2019*
Sakar English School, Ahmedabad

Secondary (10th) — *2016 - 2017*
Sakar English School, Ahmedabad

Publication

Hierarchical Deep Reinforcement Learning for S-Curve Trajectory Control and Goal-based Decision Making in Autonomous Driving

Jay Trivedi and Dr. Nitin Singh Rajput (2026). "Hierarchical Deep Reinforcement Learning for S-Curve Trajectory Control and Goal-based Decision Making in Autonomous Driving". Submitted to IEEE 34th International Conference on Software, Telecommunications, and Computer Networks (SoftCOM 2026) (Under Review)

Developed an autonomous driving system based on a reinforcement learning-driven heterogeneous hierarchical architecture, enabling efficient goal base decision-making and control in environments. The framework separates high-level strategic planning from low-level motion execution, allowing scalable and modular learning. Integrated sensor-based perception inputs to support real-time environment understanding, adaptive navigation, and lane management. The system was trained and evaluated in a simulated environment, with a focus on improving stability, sample efficiency, and robustness of learned policies. Technologies used include Python, PyTorch, Reinforcement Learning, and simulation frameworks.

Projects

Customer Support Ticket Analysis using NLP

Developed an NLP-based system for ticket classification, priority prediction, and summarization using TF-IDF features and machine learning models. Implemented Logistic Regression and SVM, performing comparative evaluation using standard metrics, with Logistic Regression achieving superior performance.

Live Bird Species Recognition System

Built a real-time bioacoustic AI system to classify 10 bird species from audio signals. Processed 5,548 recordings into log-mel spectrograms with noise filtering and segmentation. Designed a 4-layer CNN in PyTorch with batch normalization and dropout, enabling stable and efficient model training.

Air Quality Prediction using LSTM and Regression

Developed a multi-pollutant forecasting system for AQI, PM2.5, PM10, NO, NO_2, NOx, NH_3, CO, SO_2, and O_3 using CPCB data. Implemented and compared LSTM and Linear Regression models to capture temporal dynamics and baseline trends. Applied preprocessing, normalization, and feature engineering, achieving improved predictive performance on time-series data.

Supervisor Management Application

An application which gives information to any professor or volunteer about any ongoing exam for supervising. They can accept or reject the request for supervision by giving any logical reason. This application simplifies a long process.

Smart Parking System

The purpose of this project is to reduce parking rush. Easy to find parking. Smart Parking sensing technology is leading the way in the delivery of proven, fully integrated, end-to-end solutions, resulting in a transformation like Smart Sensors, SmartOHI, SmartANPR/LPR through Smart App.

Languages

English, Hindi, Gujarati